Functional Annotation of bcsA (Q88JL4 / PP_2635) in *Pseudomonas putida* KT2440

1. Summary (Answer to the Research Question)

bcsA (Q88JL4, ordered locus PP_2635) encodes the catalytic subunit of bacterial cellulose synthase (BcsA; EC 2.4.1.12). It is an integral inner-membrane, processive family-2 glycosyltransferase (GT2) that polymerizes UDP-α-D-glucose into linear β-(1→4)-glucan (cellulose), releasing UDP, and simultaneously translocates the growing polymer through its own transmembrane channel to the periplasmic/cell-surface side. Its activity is switched on by the second messenger **cyclic di-GMP (c-di-GMP)**, which binds a C-terminal PilZ domain and relieves autoinhibition. In *P. putida* KT2440, the cellulose produced by this machinery is one of several exopolysaccharides and serves as an accessory structural stabilizer of the biofilm matrix.

Gene identity was verified (see Section 2): the UniProt annotation, EC number, catalytic reaction, GT2/PilZ domain architecture, and diagnostic sequence motifs are all mutually consistent and match the well-characterized bacterial cellulose synthase family. This is the *canonical* BcsA, distinct from the recently described "orphan" BcsA-like cyclic-β-glucan synthases of pseudomonads (Section 6).

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2. Gene/Protein Identity Verification

Attribute	Value (UniProt Q88JL4)	Consistency check
Protein	Cellulose synthase catalytic subunit [UDP-forming]	✓ matches symbol *bcsA*
EC	2.4.1.12	✓ cellulose synthase (UDP-forming)
Gene / locus	*bcsA* / PP_2635	✓ matches target
Organism	*Pseudomonas putida* KT2440 (ATCC 47054 / DSM 6125)	✓ matches target
Family	Glycosyltransferase 2 (GT2)	✓
Length	869 aa	consistent with BcsA
Domains	GT2-like (356–565); PilZ (694–790); 9 TM helices	✓ canonical BcsA topology

Bioinformatic confirmation of catalytic/regulatory residues (this work): direct scan of the 869-aa sequence identified **all four** conserved catalytic motifs of processive cellulose synthases, in the canonical N→C order (Zimmer nomenclature), plus both c-di-GMP-binding motifs:

Motif	Role	Sequence found	Residues
U1 "D"	UDP/sugar binding	`TYNED`	281–285
U2 "DxD"	Mg^{2+} / UDP-glucose coordination	`FDCD`	359–362
U3 "TED"	catalytic general base (deprotonates acceptor O4)	`TED`	455–457
U4 "QxxRW"	processive glucan anchoring	`QRIRW`	493–497
PilZ RxxxR	c-di-GMP binding	`RRAHR`	(PilZ domain)
PilZ (D/N)xSxxG	c-di-GMP binding	`DYSDGG`	(PilZ domain)

The presence and correct spacing of all four catalytic motifs within the cytoplasmic GT2 domain, together with the intact PilZ c-di-GMP module, is definitive evidence of a bona fide, c-di-GMP-regulated GT-A-fold cellulose synthase — confirming we are researching the correct protein. KEGG independently assigns PP_2635 to orthology **K00694** (cellulose synthase, EC 2.4.1.12), pathway ppu00500 (starch and sucrose metabolism), with Pfam domains BcsA_N, Cellulose_synt, and PilZ.

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3. Primary Function: The Reaction Catalyzed and Substrate Specificity

BcsA is a **processive, inverting β-glycosyltransferase**. The reaction (UniProt / Rhea:19929):

> [(1→4)-β-D-glucosyl]$_n$ + UDP-α-D-glucose → [(1→4)-β-D-glucosyl]$_{(n+1)}$ + UDP + H^+ (cofactor **Mg^{2+}**)

- **Donor substrate:** UDP-α-D-glucose (the sole sugar-nucleotide donor).
- **Acceptor:** the non-reducing end of the growing β-1,4-glucan chain.
- **Product:** cellulose (linear, unbranched β-1,4-glucan).

Crystallographic and QM/MM studies of the homologous *Rhodobacter sphaeroides* BcsA–BcsB complex define the mechanism at atomic resolution: the enzyme extends the polymer **one glucose at a time** via an **S_n2-type transition state**, in which the non-reducing-end O4 hydroxyl performs nucleophilic attack on the anomeric C1 of UDP-glucose, breaking the C1–O1 glycosidic bond, with a conserved catalytic aspartate acting as general base (Morgan et al., *Nature* 2013, P 23222542; Yang et al. 2015, P 25942604). Successive glucose units are added in alternating 180° orientations to build the cellobiose repeat. Catalysis and membrane translocation are mechanistically coupled: each glucose addition ratchets the nascent chain through the BcsA transmembrane channel (Morgan et al. 2013, P 23222542).

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4. Subcellular Localization / Site of Action

- **Inner (cytoplasmic) membrane**, multi-pass integral membrane protein (UniProt; 9 predicted TM helices). The GT2 catalytic domain faces the cytoplasm (where UDP-glucose is available), while the TM helices form a **cellulose-conducting channel** that extrudes the polymer toward the periplasm (Morgan et al. 2013, P 23222542).
- Function is therefore carried out **at the inner membrane**, with the polysaccharide product delivered across it. In the complete Bcs system the periplasmic subunit **BcsB** anchors and guides the emerging chain, and additional components (e.g., BcsC, BcsZ) complete export to the cell surface (Morgan et al. 2013, P 23222542; Knecht et al. 2022, P 35289468).

The *P. putida* KT2440 *bcs* operon (genomic context, KEGG)

PP_2635 sits within a canonical, complete cellulose synthesis–secretion operon (locus ~PP_2631–PP_2638):

Locus	Gene	Product / role
PP_2631	*bcsF*/yhjT	accessory membrane protein
PP_2632	*bcsG*	putative endoglucanase / membrane accessory
PP_2634	—	putative cellulose synthase (accessory)
PP_2635	***bcsA***	**catalytic + translocator subunit (target; K00694)**
PP_2636	*bcsB*	co-catalytic, periplasmic translocator subunit
PP_2637	*bcsZ*	periplasmic endo-1,4-β-D-glucanase (chain editing)
PP_2638	*bcsC*	outer-membrane export channel (Bcs operon C protein)

This *bcsA–bcsB–bcsZ–bcsC* arrangement is the hallmark of a functional trans-envelope cellulose machine: BcsA+BcsB polymerize and translocate the glucan across the inner membrane, BcsZ trims/edits it in the periplasm, and BcsC exports it to the cell surface. The co-clustering independently corroborates the identity of PP_2635 as the cellulose-synthase catalytic subunit.

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5. Regulation and Pathway Context

- **c-di-GMP gating (allosteric activation).** BcsA's C-terminal **PilZ domain** binds c-di-GMP; binding "releases an autoinhibited state of the enzyme by breaking a salt bridge that otherwise tethers a conserved gating loop that controls access to and substrate coordination at the active site" (Morgan et al., *Nature Struct. Mol. Biol.* 2014, P 24704788). Thus cellulose synthesis is directly slaved to intracellular c-di-GMP, the master regulator of the motile→sessile (biofilm) transition.
- **Experimental validation of the PilZ motif.** In *Salmonella*, mutating the conserved **RxxxR** residues of the BcsA PilZ domain abolishes c-di-GMP-dependent cellulose production

(Zorraquino et al. 2013, P 23161026). This is direct experimental support that the very motif present in Q88JL4 (`RRAHR`) is functionally required for signal-dependent activation. That study also showed that cellulose itself acts as a physical brake on flagellar rotation at high c-di-GMP — so BcsA output reinforces the sessile lifestyle both as matrix and by steric hindrance of motility.

- **Minimal functional unit.** The authoritative review of bacterial cellulose biosynthesis (Römling & Galperin 2015, P 26077867) states that "in most studied bacteria, just two subunits, BcsA and BcsB, are necessary and sufficient for the formation of the polysaccharide chain in vitro," with auxiliary subunits modulating expression, export, and higher-order fiber organization — and with cellulose important for rhizosphere colonization and host interactions.
- **Pathway.** UniProt assigns BcsA to "Glycan metabolism; bacterial cellulose biosynthesis"; KEGG places it in ppu00500 (starch and sucrose metabolism). Cellulose production is frequently linked to biofilm formation (Morgan et al. 2014, P 24704788).
- **Regulatory network in *P. putida* KT2440.** c-di-GMP levels and downstream EPS/biofilm genes are governed by regulators such as **FleQ**, a c-di-GMP-binding transcription factor that controls flagellar and EPS operons and whose mutation reduces biofilm formation and plant-surface colonization (Molina-Henares et al. 2017, P 27503246), and by diguanylate-cyclase/phosphodiesterase enzymes (e.g., the GGDEF/EAL response regulator Rup4959, whose activity raises c-di-GMP and Calcofluor-stainable EPS while suppressing motility; Matilla et al. 2011, P 21554519). These place BcsA within the broader c-di-GMP–biofilm signaling circuitry of this organism.

6. Organism-Specific Biological Role (*P. putida* KT2440)

P. putida KT2440 possesses **four exopolysaccharide (EPS) systems** — alginate (*alg*), **cellulose (*bcs*)**, and the two novel clusters *pea* and *peb*. A dedicated genetic study (Nilsson et al. 2011, P 21507178) reports that "the gene clusters alg and bcs, which code for proteins mediating alginate and cellulose biosynthesis, were found to play minor roles in *P. putida* KT2440 biofilm formation and stability under the conditions tested," with *pea*/*peb* being the dominant matrix stabilizers. Accordingly, the primary biological output of PP_2635/BcsA — cellulose — acts as an **accessory structural stabilizer** of the biofilm rather than the principal matrix polymer in this species. This functional redundancy among EPS systems explains why a *bcs* lesion alone has limited biofilm phenotype.

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7. Supported and Refuted Hypotheses

Supported: - H1 — PP_2635/Q88JL4 is a genuine BcsA cellulose synthase catalytic subunit (EC 2.4.1.12). *Supported* by UniProt annotation + Rhea reaction + GT2 domain + diagnostic QxxRW/DxD motifs. - H2 — Activity is c-di-GMP-regulated via a C-terminal PilZ domain. *Supported* by domain annotation, presence of both RxxxR and DxSxxG c-di-GMP motifs, and structural precedent (P 24704788). - H3 — The protein acts at the inner membrane and translocates its product across it. *Supported* by 9-TM topology and BcsA–BcsB channel structure (P 23222542). - H4 — In *P. putida* KT2440 cellulose is a minor/accessory biofilm component. *Supported* by Nilsson et al. 2011 (P 21507178).

Refuted / ruled out: - The target is NOT the "orphan" BcsA-like protein that constitutes a novel pseudomonad cyclic-β-(1,3)-glucan synthase family (which shares <41% identity with true BcsA and has a distinct GH17-domain architecture; Spiers et al. 2023,

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P 37267309

Q88JL4 carries the canonical GT2 + PilZ cellulose-synthase architecture and the QxxRW/DxD cellulose-synthase motifs, so it is the *bona fide* BcsA, not an orphan. - Cellulose is NOT the dominant/essential biofilm matrix polymer in *P. putida* KT2440 (refuted by).

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8. Limitations and Future Directions

- **Direct enzymology in *P. putida* KT2440 is lacking.** The catalytic and structural evidence derives from homologs (*R. sphaeroides*, *G. hydrophila*/*E. coli*, *Komagataeibacter*). The exact substrate specificity, processivity, and c-di-GMP affinity of the *P. putida* enzyme are inferred by homology/motif conservation, not measured.
- **Cellulose chemistry.** Some pseudomonads/enterobacteria produce chemically modified cellulose (e.g., phosphoethanolamine- or acetyl-decorated). Whether *P. putida* cellulose is modified, and by which accessory *bcs* genes, was not established here.
- **Operon composition.** The full complement and arrangement of the PP_2635 *bcs* operon (BcsB/BcsC/BcsZ and any BcsE/F/G regulatory genes) in KT2440 warrants explicit mapping.
- **Conditional phenotypes.** The "minor" biofilm role was established under specific laboratory conditions; cellulose may be more important during plant-root/rhizosphere colonization, desiccation resistance, or phage defense (cf. cellulose as a phage receptor in *Erwinia*,).

  P 35289468

References (PMIDs)

- **26077867** — Römling, Galperin. Bacterial cellulose biosynthesis: diversity of operons, subunits, products, and functions. *Trends Microbiol.* (2015). *(authoritative review)*
- **23161026** — Zorraquino et al. Coordinated cyclic-di-GMP repression of *Salmonella* motility through YcgR and cellulose. *J. Bacteriol.* (2013). *(RxxxR PilZ mutagenesis)*

- **23222542** — Morgan, Strumillo, Zimmer. Crystallographic snapshot of cellulose synthesis and membrane translocation. *Nature* (2013).
- **24704788** — Morgan, McNamara, Zimmer. Mechanism of activation of bacterial cellulose synthase by cyclic di-GMP. *Nat. Struct. Mol. Biol.* (2014).
- **25942604** — Yang, Zimmer, Yingling, Kubicki. QM/MM study of cellulose polymerization mechanism in bacterial CESA (2015).
- **21507178** — Nilsson et al. Influence of putative exopolysaccharide genes on *P. putida* KT2440 biofilm stability (2011).
- **27503246** — Molina-Henares et al. FleQ of *P. putida* KT2440 is a multimeric c-di-GMP-binding regulator of biofilm matrix components (2017).
- **21554519** — Matilla et al. Cyclic diguanylate turnover by the sole GGDEF/EAL response regulator in *P. putida* (2011).
- **35289468** — Knecht et al. Bacteriophage S6 requires bacterial cellulose for *Erwinia amylovora* infection (2022).
- **37267309** — Spiers et al. Bioinformatics characterization of BcsA-like orphan proteins suggests a novel family of pseudomonad cyclic-β-glucan synthases (2023) — used to *exclude* misidentification.
- UniProt **Q88JL4**; Rhea **RHEA:19929**; InterPro IPR003919, IPR001173, IPR009875.
